

feature	prerequisites	test case	procedure	data	expected result	actual result	status	comments
Signup	The server is running.	Verify the success of the signup process and return the correct status code when the payload contains all the correct data.	1.Open postman 2.Put the method to POST 3.Put the endpoint http://localhost:8080/signup	{ "username": "testuser", "password": "testpassword", "phone": "1234567890" }	The user must be added successfully, and the status code must be 201 (Created).	The user was successfully added; the response is (success:"true"), and the status code is 201 (Created).	pass	
	The server is running. A user with the same username or phone already exists.	Verify that signup fails when username or phone number already exists.	1.Open postman 2.Put the method to POST 3.Put the endpoint http://localhost:8080/signup	{ "username": "testuser", "password": "testpassword", "phone": "1234567890" }	The user must not be added; the status code should be 409 (Conflict), and an error message must appear in the response.	The user was not added; and the status code is 401 (Unauthorized) with the following reply message: { "success": false, "message": "Username or phone number already exists"	fail	This is a bug, because the requirements specify that the status code must be 409.
	The server is running.	Verify signup fails when username field is empty	1.Open postman 2.Put the method to POST 3.Put the endpoint http://localhost:8080/signup 4.Keep the username empty	{ "username": "", "password": "testpassword", "phone": "1234567890" }	The user must not be added; the status code should be 204 (Empty), and an error message must appear in the response.	The user was successfully added; the response is (success:"true"), and the status code is 201 (Created).	fail	This is a bug; the user should not be add, and the response status code should be 204.
	The server is running.	Verify signup fails when password field is empty	1.Open postman 2.Put the method to POST 3.Put the endpoint http://localhost:8080/signup 4.Keep the password empty	{ "username": "testuser", "password": "", "phone": "1234567890" }	The user must not be added; the status code should be 204 (Empty), and an error message must appear in the response.	The user was successfully added; the response is (success:"true"), and the status code is 201 (Created).	fail	This is a bug; the user should not be add, and the response status code should be 204.
	The server is running.	Verify signup fails when phone field is empty	1.Open postman 2.Put the method to POST 3.Put the endpoint http://localhost:8080/signup 4.Keep the phone empty	{ "username": "testuser", "password": "testpassword", "phone": "" }	The user must not be added; the status code should be 204 (Empty), and an error message must appear in the response.	The user was successfully added; the response is (success:"true"), and the status code is 201 (Created).	fail	This is a bug; the user should not be add, and the response status code should be 204.
login	The server is running.	Verify signup fails when fields contain only whitespace characters.	1.Open postman 2.Put the method to POST 3.Put the endpoint http://localhost:8080/signup 4.Enter space in all fields	{ "username": " ", "password": " ", "phone": " " }	The user must not be added; the status code should be 400 (Bad Request), and an error message must appear in the response.	The user was successfully added; the response is (success:"true"), and the status code is 201 (Created).	fail	This is a bug; the user should not be add, and the response status code should be 400.
	The server is running.	Verify signup fails when phone contains characters or special symbols.	1.Open postman 2.Put the method to POST 3.Put the endpoint http://localhost:8080/signup	{ "username": "testuser", "password": "testpassword", "phone": "&&aae" }	The user must not be added; the status code should be 400 (Bad Request), and an error message must appear in the response.	The user was successfully added; the response is (success:"true"), and the status code is 201 (Created).	fail	This is a bug; the user should not be add, and the response status code should be 400.
	The server is running. The user has signed up	Verify that the correct status code is returned when logging in with a valid username and password.	1.Open postman 2.Put the method to POST 3.Put the endpoint http://localhost:8080/login	{ "username": "testuser", "password": "testpassword", }	The user must be logged in successfully, and the status code must be 200 (OK).	The user was successfully logged in; the response is (success:"true"), and the status code is 201 (Created).	fail	This is a bug; response status code should be 200.
	The server is running. The user has signed up	Verify that the correct status code is returned when logging in with a invalid username and valid password.	1.Open postman 2.Put the method to POST 3.Put the endpoint http://localhost:8080/login	{ "username": "testuser12", "password": "testpassword", }	The user must not be logged in successfully, and the status code must be 401 (Unauthorized).	The user wasn't successfully logged in; the response is { "success": false, "message": "Invalid username or password" }, and the status code is 401 (Unauthorized).	pass	
	The server is running. The user has signed up	Verify that the correct status code is returned when logging in with a valid username and invalid password.	1.Open postman 2.Put the method to POST 3.Put the endpoint http://localhost:8080/login	{ "username": "testuser", "password": "qwqwqwqw", }	The user must not be logged in successfully, and the status code must be 401 (Unauthorized).	The user wasn't successfully logged in; the response is { "success": false, "message": "Invalid username or password" }, and the status code is 401 (Unauthorized).	pass	
Book	The server is running. The user has signed up The user has logged in	Verify successful appointment booking when valid future date and time are provided.	1.Open postman 2.Put the method to POST 3.Put the endpoint http://localhost:8080/book	{ "date": "2027-08-15", "startTime": "10:00", "endTime": "11:00" }	The appointment must be booked successfully, and the status code must be 201 (Created).	The appointment was successfully booked; the response is (success:"true"), and the status code is 201 (Created).	pass	
	The server is running. The user has signed up The user has logged in	Verify booking fails when attempting to book an appointment with a past date.	1.Open postman 2.Put the method to POST 3.Put the endpoint http://localhost:8080/book	{ "date": "2025-08-15", "startTime": "10:00", "endTime": "11:00" }	The appointment must not be booked, and the status code must be 400 (Bad Request).	The appointment wasn't booked; the response is { "success": false, "message": "Cannot book an appointment in the past" } and the status code is 409 (Conflict).	fail	This is a bug; response status code should be 400.
	The server is running. The user has signed up The user has logged in	Verify booking fails when appointment duration is less than 30 minutes.	1.Open postman 2.Put the method to POST 3.Put the endpoint http://localhost:8080/book	{ "date": "2027-08-15", "startTime": "10:00", "endTime": "10:29" }	The appointment must not be booked, and the status code must be 400 (Bad Request).	The appointment was successfully booked; the response is (success:"true"), and the status code is 201 (Created).	fail	This is a bug; The appointment must not be booked response status code should be 400.
	The server is running. The user has signed up The user has logged in	Verify booking succeeds when duration is more than 30 minutes.	1.Open postman 2.Put the method to POST 3.Put the endpoint http://localhost:8080/book	{ "date": "2027-08-15", "startTime": "10:00", "endTime": "11:00" }	The appointment must be booked successfully, and the status code must be 201 (Created).	The appointment was successfully booked; the response is (success:"true"), and the status code is 201 (Created).	pass	
	The server is running. The user has signed up The user has logged in	Verify booking succeeds when duration is exactly 30 minutes.	1.Open postman 2.Put the method to POST 3.Put the endpoint http://localhost:8080/book	{ "date": "2027-08-15", "startTime": "10:00", "endTime": "10:30" }	The appointment must be booked successfully, and the status code must be 201 (Created).	The appointment was successfully booked; the response is (success:"true"), and the status code is 201 (Created).	pass	
	The server is running. The user has signed up The user has logged in	Verify booking fails when end time is before start time.	1.Open postman 2.Put the method to POST 3.Put the endpoint http://localhost:8080/book	{ "date": "2027-08-15", "startTime": "10:00", "endTime": "9:30" }	The appointment must not be booked, and the status code must be 400 (Bad Request).	The appointment wasn't booked; the response is { "success": false, "message": "Appointment must be at least 20 minutes" } and the status code is 400 (Bad Request).	fail	This is a bug;
	The server is running. The user has signed up The user has logged in	Verify booking fails when appointment completely overlaps an existing booking.	1.Open postman 2.Put the method to POST 3.Put the endpoint http://localhost:8080/book	{ "date": "2027-08-15", "startTime": "10:00", "endTime": "11:00" }	The appointment must not be booked, and the status code must be 409 (Conflict).	The appointment wasn't booked; the response is { "success": false, "message": "Appointment slot already booked or overlaps with another appointment" } and the status code is 409 (Conflict).	pass	
	The server is running. The user has signed up The user has logged in	Verify booking succeeds when start time equals previous appointment end time.	1.Open postman 2.Put the method to POST 3.Put the endpoint http://localhost:8080/book	{ "date": "2027-08-15", "startTime": "11:00", "endTime": "12:00" }	The appointment must be booked successfully, and the status code must be 201 (Created).	The appointment was successfully booked; the response is (success:"true"), and the status code is 201 (Created).	pass	
	The server is running. The user has signed up The user has logged in	Verify booking fails when new appointment partially overlaps an existing appointment.	1.Open postman 2.Put the method to POST 3.Put the endpoint http://localhost:8080/book	{ "date": "2027-08-15", "startTime": "10:10", "endTime": "11:10" }	The appointment must not be booked, and the status code must be 409 (Conflict).	The appointment wasn't booked; the response is { "success": false, "message": "Appointment slot already booked or overlaps with another appointment" } and the status code is 409 (Conflict).	pass	
	The server is running. The user has signed up The user has logged in	Verify booking fails when overlap occurs by one minute.	1.Open postman 2.Put the method to POST 3.Put the endpoint http://localhost:8080/book	{ "date": "2027-08-15", "startTime": "10:59", "endTime": "12:00" }	The appointment must not be booked, and the status code must be 409 (Conflict).	The appointment wasn't booked; the response is { "success": false, "message": "Appointment slot already booked or overlaps with another appointment" } and the status code is 409 (Conflict).	pass	

	The server is running. The user has signed up The user has logged in	Verify booking fails when date is missing.	1.Open postman 2.Put the method to POST 3.Put the endpoint http://localhost:8080/book 4.Keep the date empty	{ "date": "", "startTime": "10:59", "endTime": "12:00" }	The appointment must not be booked, and the status code must be 400 (Bad Request).	The appointment wasn't booked; the response is { "success": false, "message": "Cannot book an appointment in the past" } and the status code is 409 (Conflict).	fail	This is a Bug. The status code should be 400, and the response message should be: Cannot keep an empty field.
	The server is running. The user has signed up The user has logged in	Verify booking fails when startTime is missing.	1.Open postman 2.Put the method to POST 3.Put the endpoint http://localhost:8080/book 4.Keep the startTime empty	{ "date": "2027-08-15", "startTime": "", "endTime": "12:00" }	The appointment must not be booked, and the status code must be 400 (Bad Request).	The appointment wasn't booked; the response is { "success": false, "message": "Cannot book an appointment in the past" } and the status code is 409 (Conflict).	fail	This is a Bug. The status code should be 400, and the response message should be: Cannot keep an empty field.
	The server is running. The user has signed up The user has logged in	Verify booking fails when endTime is missing.	1.Open postman 2.Put the method to POST 3.Put the endpoint http://localhost:8080/book 4.Keep the endTime empty	{ "date": "2027-08-15", "startTime": "10:59", "endTime": "" }	The appointment must not be booked, and the status code must be 400 (Bad Request).	The appointment wasn't booked; the response is { "success": false, "message": "Appointment must be at least 20 minutes" } and the status code is 400 (Bad Request).	fail	This is a Bug. the response message should be: Cannot keep an empty field.
	The server is running. The user has signed up The user has logged in	Verify booking fails when invalid date format is provided.	1.Open postman 2.Put the method to POST 3.Put the endpoint http://localhost:8080/book	{ "date": "15/08/2027", "startTime": "10:59", "endTime": "12:00" }	The appointment must not be booked, and the status code must be 400 (Bad Request).	The appointment wasn't booked; the response is { "success": false, "message": "Cannot book an appointment in the past" } and the status code is 409 (Conflict).	fail	This is a Bug. The status code should be 400, and the response message should be: invalid date data.
	The server is running. The user has signed up The user has logged in	Verify booking fails when invalid startTime format is provided.	1.Open postman 2.Put the method to POST 3.Put the endpoint http://localhost:8080/book	{ "date": "2027-08-15", "startTime": "10:59", "endTime": "12:00" }	The appointment must not be booked, and the status code must be 400 (Bad Request).	The appointment wasn't booked; the response is { "success": false, "message": "Cannot book an appointment in the past" } and the status code is 409 (Conflict).	fail	This is a Bug. The status code should be 400, and the response message should be: invalid date data.
	The server is running. The user has signed up The user has logged in	Verify booking fails when invalid endTime format is provided.	1.Open postman 2.Put the method to POST 3.Put the endpoint http://localhost:8080/book	{ "date": "2027-08-15", "startTime": "10:59", "endTime": "qweq-qwe" }	The appointment must not be booked, and the status code must be 400 (Bad Request).	The appointment wasn't booked; the response is { "success": false, "message": "Cannot book an appointment in the past" } and the status code is 409 (Conflict).	fail	This is a Bug. The status code should be 400, and the response message should be: invalid date data.
cancel book	The server is running. The user has signed up The user has logged in Existing appointment	Verify successful appointment cancellation.	1.Open postman 2.Put the method to POST 3.Put the endpoint http://localhost:8080/cancel	{ "date": "2027-08-15", "startTime": "10:59", "endTime": "qweq-qwe", "username": "khader" }	The appointment should be successfully canceled and the status code should be 200 (OK)	The appointment canceled successfully and the response { "success": true } and the status code is 200 (OK)	pass	
	The server is running. The user has signed up The user has logged in Existing appointment	Verify cancellation fails when one or more required fields are missing.	1.Open postman 2.Put the method to POST 3.Put the endpoint http://localhost:8080/cancel 4.Keep one of field empty	{ "date": "", "startTime": "", "endTime": "", "username": "" }	The appointment must not be booked, and the status code must be 400 (Bad Request).	The appointment doesn't canceled successfully and the response { "success": false, "message": "Appointment canceled successfully" } and the status code is 401 (Unauthorized)	fail	This is a Bug
	The server is running. The user has signed up The user has logged in Existing appointment	Verify cancellation fails when appointment does not exist.	1.Open postman 2.Put the method to POST 3.Put the endpoint http://localhost:8080/cancel	{ "date": "2027-08-11", "startTime": "10:59", "endTime": "qweq-qwe", "username": "khader" }	The appointment must not be booked, and the status code must be 404 (Not Found).	The appointment doesn't canceled successfully and the response { "success": false, "message": "Appointment canceled successfully" } and the status code is 401 (Unauthorized)	fail	This is a Bug
delete user	The server is running. The user has signed up The user has logged in	Make sure there is an endpoint for deleting the user.	1.Open postman 2.Put the method to POST 3.Put the endpoint http://localhost:8080/delete		When the endpoint is placed with the username, it must be deleted.	There is no endpoint to that	fail	This is a Bug